Rami L.

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EDUCATION

Bachelor of Science, Electrical & Computer Engineering Engineering, Graduating Spring 2028
The University of Texas at Austin
Relevant Coursework: Digital Logic Design, Software Design and Implementation I, Intro to Embedded Systems, Circuit Theory, Linear Systems and Signals, Calculus 1-3, Engineering Physics 1-2
Misc. Clubs and Activities: Texas Longhorn Band Piccolo, Kappa Kappa Psi Service Fraternity Active Member, Filipino Student Association Member, Institute of Electrical and Electronics Engineers

WORK EXPERIENCE

Incoming Embedded Hardware Intern, Clairvoyant Networks, Summer 2026
Radio Systems Engineer, Longhorn Lemons, 2025-2026

- Reviewed preliminary circuit design concepts for a student-built endurance race car, focusing on wiring layout, power distribution, and system reliability
- Contributed to early electrical planning discussions with mechanical and design subteams
- Gained exposure to automotive electrical design constraints including durability, safety, and maintainability

PROJECTS

Space Invaders Game System, ECE 319k (Intro to Embedded Systems), Spring 2025

- Interfaced, wired, and programmed a full game system from scratch with pressable switches, a slide pot joystick functioning via analog-digital conversion (ADC), a speaker with audio files processed via digital-analog conversion (DAC), and an LCD display wired to the Texas Instruments MSPM0 microcontroller
- Programmed a working game start and end screen with two language options (English and Spanish), and implemented game sprites in the C language with ST7735R TFT drivers

Programmable Stopwatch/Timer, ECE 316 (Digital Logic Design), Spring 2026

- Designed a programmable stopwatch/timer on a Basys3 FPGA using Verilog RTL, supporting count-up/count-down operation and externally loaded preset values
- Built a controller/datapath architecture with HLSM-based design planning, mode selection logic, start/stop control, reset behavior, and 10 ms timing resolution across four 7-segment displays
- Verified full hardware functionality in Vivado and on the Basys3 board using buttons, switches, clocking logic, constraints, and generated bitstream files

Octave-spaced FIR Filter Banks - ECE 313 (Linear Systems and Signals), Fall 2025

- Designed and implemented a four-level digital signal processing filter bank to decompose audio signals into five octave-spaced frequency bands using MATLAB
- Analyzed multirate signal processing systems by deriving effective frequency responses for cascaded FIR filters with downsampling and upsampling operations
- Achieved near-perfect reconstruction (PSNR >18 dB) while demonstrating compression ratios up to 1/16 for audio compression applications

TECHNICAL SKILLS AND CERTIFICATIONS

Programming Knowledge: C/C++, Python, Java, HTML, CSS, Verilog, LC3/ARM Assembly
Computer Skills: Excel/Google Sheets, Visual Studio Code, Github, Xilinx Vivado, KiCAD, MATLAB